

Chapter 3

Maximum likelihood and Bayesian parameter estimation

3.1 Introduction

In Chap. ?? we saw how we could design an optimal classifier if we knew the prior probabilities $P(\omega_i)$ and the class-conditional densities $p(\mathbf{x}|\omega_i)$. Unfortunately, in pattern recognition applications we rarely if ever have this kind of complete knowledge about the probabilistic structure of the problem. In a typical case we merely have some vague, general knowledge about the situation, together with a number of *design samples* or *training data* — particular representatives of the patterns we want to classify. The problem, then, is to find some way to use this information to design or train the classifier.

TRAINING
DATA

One approach to this problem is to use the samples to estimate the unknown probabilities and probability densities, and to use the resulting estimates as if they were the true values. In typical supervised pattern classification problems, the estimation of the prior probabilities presents no serious difficulties (Problem 3). However, estimation of the class-conditional densities is quite another matter. The number of available samples always seems too small, and serious problems arise when the dimensionality of the feature vector $\mathbf{x}$ is large. If we know the number of parameters in advance and our general knowledge about the problem permits us to parameterize the conditional densities, then the severity of these problems can be reduced significantly. Suppose, for example, that we can reasonably assume that $p(\mathbf{x}|\omega_i)$ is a normal density with mean $\boldsymbol{\mu}_i$ and covariance matrix $\boldsymbol{\Sigma}_i$, although we do not know the exact values of these quantities. This knowledge simplifies the problem from one of estimating an unknown *function* $p(\mathbf{x}|\omega_i)$ to one of estimating the *parameters* $\boldsymbol{\mu}_i$ and $\boldsymbol{\Sigma}_i$.

The problem of parameter estimation is a classical one in statistics, and it can be approached in several ways. We shall consider two common and reasonable procedures, *maximum likelihood* estimation and *Bayesian* estimation. Although the results obtained with these two procedures are frequently nearly identical, the approaches

MAXIMUM
LIKELIHOOD

BAYESIAN
ESTIMATION

are conceptually quite different. Maximum likelihood and several other methods view the parameters as quantities whose values are fixed but unknown. The best estimate of their value is defined to be the one that maximizes the probability of obtaining the samples actually observed. In contrast, Bayesian methods view the parameters as random variables having some known a priori distribution. Observation of the samples converts this to a posterior density, thereby revising our opinion about the true values of the parameters. In the Bayesian case, we shall see that a typical effect of observing additional samples is to sharpen the a posteriori density function, causing it to peak near the true values of the parameters. This phenomenon is known as *Bayesian learning*. In either case, we use the posterior densities for our classification rule, as we have seen before.

It is important to distinguish between supervised learning and unsupervised learning. In both cases, samples $\mathbf{x}$ are assumed to be obtained by selecting a state of nature ω_i with probability $P(\omega_i)$, and then independently selecting $\mathbf{x}$ according to the probability law $p(\mathbf{x}|\omega_i)$. The distinction is that with supervised learning we know the state of nature (class label) for each sample, whereas with unsupervised learning we do not. As one would expect, the problem of unsupervised learning is the more difficult one. In this chapter we shall consider only the supervised case, deferring consideration of unsupervised learning to Chap. ??.

3.2 Maximum Likelihood Estimation

Maximum likelihood estimation methods have a number of attractive attributes. First, they nearly always have good convergence properties as the number of training samples increases. Further, maximum likelihood estimation often can be simpler than alternate methods, such as Bayesian techniques or other methods presented in subsequent chapters.

3.2.1 The General Principle

Suppose that we separate a collection of samples according to class, so that we have c sets, $\mathcal{D}_1, \dots, \mathcal{D}_c$, with the samples in $\mathcal{D}_j$ having been drawn independently according to the probability law $p(\mathbf{x}|\omega_j)$. We say such samples are *i.i.d.* — independent identically distributed random variables. We assume that $p(\mathbf{x}|\omega_j)$ has a known parametric form, and is therefore determined uniquely by the value of a parameter vector θ_j . For example, we might have $p(\mathbf{x}|\omega_j) \sim N(\mu_j, \Sigma_j)$, where θ_j consists of the components of μ_j and Σ_j . To show the dependence of $p(\mathbf{x}|\omega_j)$ on θ_j explicitly, we write $p(\mathbf{x}|\omega_j)$ as $p(\mathbf{x}|\omega_j, \theta_j)$. Our problem is to use the information provided by the training samples to obtain good estimates for the unknown parameter vectors $\theta_1, \dots, \theta_c$ associated with each category.

To simplify treatment of this problem, we shall assume that samples in $\mathcal{D}_i$ give no information about θ_j if $i \neq j$ — that is, we shall assume that the parameters for the different classes are functionally independent. This permits us to work with each class separately, and to simplify our notation by deleting indications of class distinctions. With this assumption we thus have c separate problems of the following form: Use a set $\mathcal{D}$ of training samples drawn independently from the probability density $p(\mathbf{x}|\theta)$ to estimate the unknown parameter vector θ .

Suppose that $\mathcal{D}$ contains n samples, $\mathbf{x}_1, \dots, \mathbf{x}_n$. Then, since the samples were drawn independently, we have

$$p(\mathcal{D}|\boldsymbol{\theta}) = \prod_{k=1}^n p(\mathbf{x}_k|\boldsymbol{\theta}). \quad (1)$$

Recall from Chap. ?? that, viewed as a function of $\boldsymbol{\theta}$, $p(\mathcal{D}|\boldsymbol{\theta})$ is called the *likelihood* of $\boldsymbol{\theta}$ with respect to the set of samples. The *maximum likelihood estimate* of $\boldsymbol{\theta}$ is, by definition, the value $\hat{\boldsymbol{\theta}}$ that maximizes $p(\mathcal{D}|\boldsymbol{\theta})$. Intuitively, this estimate corresponds to the value of $\boldsymbol{\theta}$ that in some sense best agrees with or supports the actually observed training samples (Fig. 3.1).

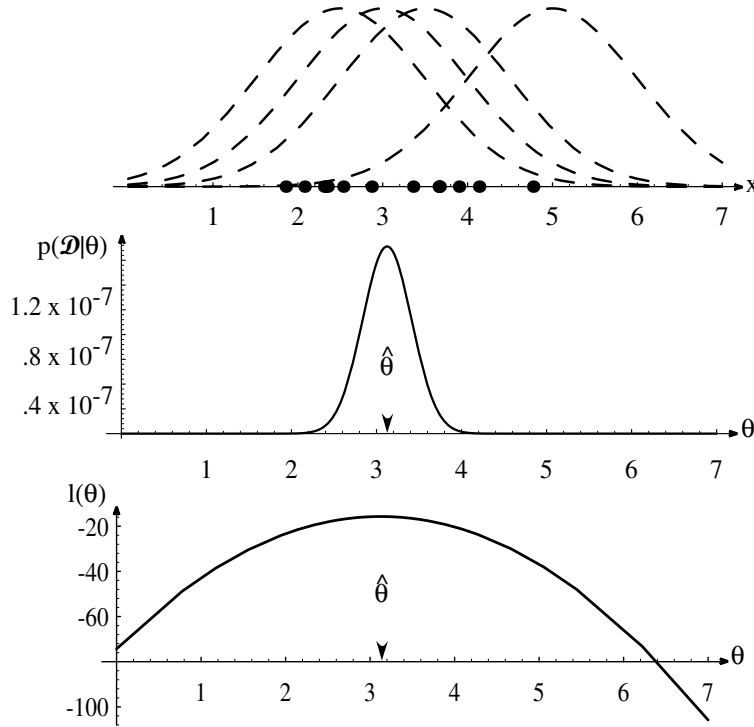


Figure 3.1: The top graph shows several training points in one dimension, known or assumed to be drawn from a Gaussian of a particular variance, but unknown mean. Four of the infinite number of candidate source distributions are shown in dashed lines. The middle figures shows the likelihood $p(\mathcal{D}|\theta)$ as a function of the mean. If we had a very large number of training points, this likelihood would be very narrow. The value that maximizes the likelihood is marked $\hat{\theta}$; it also maximizes the logarithm of the likelihood — i.e., the log-likelihood $l(\theta)$, shown at the bottom. Note especially that the likelihood lies in a different space from $p(x|\hat{\theta})$, and the two can have different functional forms.

For analytical purposes, it is usually easier to work with the logarithm of the likelihood than with the likelihood itself. Since the logarithm is monotonically increasing, the $\hat{\boldsymbol{\theta}}$ that maximizes the log-likelihood also maximizes the likelihood. If $p(\mathcal{D}|\boldsymbol{\theta})$ is a well behaved, differentiable function of $\boldsymbol{\theta}$, $\hat{\boldsymbol{\theta}}$ can be found by the standard methods of differential calculus. If the number of parameters to be set is p , then we let $\boldsymbol{\theta}$ denote

the p -component vector $\boldsymbol{\theta} = (\theta_1, \dots, \theta_p)^t$, and $\nabla_{\boldsymbol{\theta}}$ be the gradient operator

$$\nabla_{\boldsymbol{\theta}} \equiv \begin{bmatrix} \frac{\partial}{\partial \theta_1} \\ \vdots \\ \frac{\partial}{\partial \theta_p} \end{bmatrix}. \quad (2)$$

LOG-
LIKELIHOOD

We define $l(\boldsymbol{\theta})$ as the *log-likelihood* function*

$$l(\boldsymbol{\theta}) \equiv \ln p(\mathcal{D}|\boldsymbol{\theta}). \quad (3)$$

We can then write our solution formally as the argument $\boldsymbol{\theta}$ that maximizes the log-likelihood, i.e.,

$$\hat{\boldsymbol{\theta}} = \arg \max_{\boldsymbol{\theta}} l(\boldsymbol{\theta}), \quad (4)$$

where the dependence on the data set $\mathcal{D}$ is implicit. Thus we have from Eq. 1

$$l(\boldsymbol{\theta}) = \sum_{k=1}^n \ln p(\mathbf{x}_k|\boldsymbol{\theta}) \quad (5)$$

and

$$\nabla_{\boldsymbol{\theta}} l = \sum_{k=1}^n \nabla_{\boldsymbol{\theta}} \ln p(\mathbf{x}_k|\boldsymbol{\theta}). \quad (6)$$

Thus, a set of necessary conditions for the maximum likelihood estimate for $\boldsymbol{\theta}$ can be obtained from the set of p equations

$$\boxed{\nabla_{\boldsymbol{\theta}} l = \mathbf{0}.} \quad (7)$$

A solution $\hat{\boldsymbol{\theta}}$ to Eq. 7 could represent a true global maximum, a *local* maximum or minimum, or (rarely) an inflection point of $l(\boldsymbol{\theta})$. One must be careful, too, to check if the extremum occurs at a boundary of the parameter space, which might not be apparent from the solution to Eq. 7. If all solutions are found, we are guaranteed that one represents the true maximum, though we might have to check each solution individually (or calculate second derivatives) to identify which is the global optimum. Of course, we must bear in mind that $\hat{\boldsymbol{\theta}}$ is an estimate; it is only in the limit of an infinitely large number of training points that we can expect that our estimate will equal to the true value of the generating function (Sec. 3.5.1).

MAXIMUM A
POSTERIORI

MODE

We note in passing that a related class of estimators — *maximum a posteriori* or MAP estimators — find the value of $\boldsymbol{\theta}$ that maximizes $l(\boldsymbol{\theta})p(\boldsymbol{\theta})$. Thus a maximum likelihood estimator is a MAP estimator for the uniform or “flat” prior. As such, a MAP estimator finds the peak, or *mode* of a posterior density. The drawback of MAP estimators is that if we choose some arbitrary nonlinear transformation of the parameter space (e.g., an overall rotation), the density will change, and our MAP solution need no longer be appropriate (Sec. 3.5.2).

* Of course, the base of the logarithm can be chosen for convenience, and in most analytic problems base e is most natural. For that reason we will generally use $\ln$ rather than $\log$ or $\log_2$.

3.2.2 The Gaussian Case: Unknown μ

To see how maximum likelihood methods results apply to a specific case, suppose that the samples are drawn from a multivariate normal population with mean μ and covariance matrix Σ . For simplicity, consider first the case where only the mean is unknown. Under this condition, we consider a sample point $\mathbf{x}_k$ and find

$$\ln p(\mathbf{x}_k|\mu) = -\frac{1}{2}\ln[(2\pi)^d|\Sigma|] - \frac{1}{2}(\mathbf{x}_k - \mu)^t \Sigma^{-1}(\mathbf{x}_k - \mu) \quad (8)$$

and

$$\nabla_{\theta} \ln p(\mathbf{x}_k|\mu) = \Sigma^{-1}(\mathbf{x}_k - \mu). \quad (9)$$

Identifying θ with μ , we see from Eq. 9 that the maximum likelihood estimate for μ must satisfy

$$\sum_{k=1}^n \Sigma^{-1}(\mathbf{x}_k - \hat{\mu}) = \mathbf{0}, \quad (10)$$

that is, each of the d components of $\hat{\mu}$ must vanish. Multiplying by Σ and rearranging, we obtain

$$\hat{\mu} = \frac{1}{n} \sum_{k=1}^n \mathbf{x}_k. \quad (11)$$

This is a very satisfying result. It says that the maximum likelihood estimate for the unknown population mean is just the arithmetic average of the training samples — the *sample mean*, sometimes written $\hat{\mu}_n$ to clarify its dependence on the number of samples. Geometrically, if we think of the n samples as a cloud of points, the sample mean is the centroid of the cloud. The sample mean has a number of desirable statistical properties as well, and one would be inclined to use this rather obvious estimate even without knowing that it is the maximum likelihood solution.

SAMPLE
MEAN

3.2.3 The Gaussian Case: Unknown μ and Σ

In the more general (and more typical) multivariate normal case, neither the mean μ nor the covariance matrix Σ is known. Thus, these unknown parameters constitute the components of the parameter vector θ . Consider first the univariate case with $\theta_1 = \mu$ and $\theta_2 = \sigma^2$. Here the log-likelihood of a single point is

$$\ln p(x_k|\theta) = -\frac{1}{2} \ln 2\pi\theta_2 - \frac{1}{2\theta_2}(x_k - \theta_1)^2 \quad (12)$$

and its derivative is

$$\nabla_{\theta} l = \nabla_{\theta} \ln p(x_k|\theta) = \begin{bmatrix} \frac{1}{\theta_2}(x_k - \theta_1) \\ -\frac{1}{2\theta_2} + \frac{(x_k - \theta_1)^2}{2\theta_2^2} \end{bmatrix}. \quad (13)$$

Applying Eq. 7 to the full log-likelihood leads to the conditions

$$\sum_{k=1}^n \frac{1}{\hat{\theta}_2}(x_k - \hat{\theta}_1) = 0 \quad (14)$$

and

$$-\sum_{k=1}^n \frac{1}{\hat{\theta}_2} + \sum_{k=1}^n \frac{(x_k - \hat{\theta}_1)^2}{\hat{\theta}_2^2} = 0, \quad (15)$$

where $\hat{\theta}_1$ and $\hat{\theta}_2$ are the maximum likelihood estimates for θ_1 and θ_2 , respectively. By substituting $\hat{\mu} = \hat{\theta}_1$, $\hat{\sigma}^2 = \hat{\theta}_2$ and doing a little rearranging, we obtain the following maximum likelihood estimates for μ and σ^2 :

$$\hat{\mu} = \frac{1}{n} \sum_{k=1}^n x_k \quad (16)$$

and

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{k=1}^n (x_k - \hat{\mu})^2. \quad (17)$$

While the analysis of the multivariate case is basically very similar, considerably more manipulations are involved (Problem 6). Just as we would predict, though, the result is that the maximum likelihood estimates for $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ are given by

$$\hat{\boldsymbol{\mu}} = \frac{1}{n} \sum_{k=1}^n \mathbf{x}_k \quad (18)$$

and

$$\hat{\boldsymbol{\Sigma}} = \frac{1}{n} \sum_{k=1}^n (\mathbf{x}_k - \hat{\boldsymbol{\mu}})(\mathbf{x}_k - \hat{\boldsymbol{\mu}})^t. \quad (19)$$

Thus, once again we find that the maximum likelihood estimate for the mean vector is the sample mean. The maximum likelihood estimate for the covariance matrix is the arithmetic average of the n matrices $(\mathbf{x}_k - \hat{\boldsymbol{\mu}})(\mathbf{x}_k - \hat{\boldsymbol{\mu}})^t$. Since the true covariance matrix is the expected value of the matrix $(\mathbf{x} - \hat{\boldsymbol{\mu}})(\mathbf{x} - \hat{\boldsymbol{\mu}})^t$, this is also a very satisfying result.

3.2.4 Bias

BIAS

The maximum likelihood estimate for the variance σ^2 is *biased*; that is, the expected value over all data sets of size n of the sample variance is not equal to the true variance:*

$$\mathcal{E} \left[\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2 \right] = \frac{n-1}{n} \sigma^2 \neq \sigma^2. \quad (20)$$

We shall return to a more general consideration of bias in Chap. ??, but for the moment we can verify Eq. 20 for an underlying distribution with non-zero variance, σ^2 , in the extreme case of $n = 1$, in which the expectation value $\mathcal{E}[\cdot] = 0 \neq \sigma^2$. The maximum likelihood estimate of the covariance matrix is similarly biased.

Elementary *unbiased* estimators for σ^2 and $\boldsymbol{\Sigma}$ are given by

* There should be no confusion over this use of the *statistical* term bias, and that for an offset in neural networks and many other places.

$$\mathcal{E} \left[\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 \right] = \sigma^2 \quad \text{and} \quad (21)$$

$$\mathbf{C} = \frac{1}{n-1} \sum_{k=1}^n (\mathbf{x}_k - \hat{\boldsymbol{\mu}})(\mathbf{x}_k - \hat{\boldsymbol{\mu}})^t, \quad (22)$$

where $\mathbf{C}$ is the so-called *sample covariance matrix*, as explored in Problem 33. If an estimator is unbiased for *all* distributions, as for example the variance estimator in Eq. 21, then it is called *absolutely unbiased*. If the estimator tends to become unbiased as the number of samples becomes very large, as for instance Eq. 20, then the estimator is *asymptotically unbiased*. In many pattern recognition problems with large training data sets, asymptotically unbiased estimators are acceptable.

Clearly, $\hat{\boldsymbol{\Sigma}} = [(n-1)/n]\mathbf{C}$, and $\hat{\boldsymbol{\Sigma}}$ is asymptotically unbiased — these two estimates are essentially identical when n is large. However, the existence of two similar but nevertheless distinct estimates for the covariance matrix may be disconcerting, and it is natural to ask which one is “correct.” Of course, for $n > 1$ the answer is that these estimates are neither right nor wrong — they are just different. What the existence of two actually shows is that no single estimate possesses all of the properties we might desire. For our purposes, the most desirable property is rather complex — we want the estimate that leads to the best classification performance. While it is usually both reasonable and sound to design a classifier by substituting the maximum likelihood estimates for the unknown parameters, we might well wonder if other estimates might not lead to better performance. Below we address this question from a Bayesian viewpoint.

If we have a reliable model for the underlying distributions and their dependence upon the parameter vector $\boldsymbol{\theta}$, the maximum likelihood classifier will give excellent results. But what if our model is wrong — do we nevertheless get the best classifier in our assumed set of models? For instance, what if we assume that a distribution comes from $N(\mu, 1)$ but instead it actually comes from $N(\mu, 10)$? Will the value we find for $\theta = \mu$ by maximum likelihood yield the best of all classifiers of the form derived from $N(\mu, 1)$? Unfortunately, the answer is “no,” and an illustrative counterexample is given in Problem 7 where the so-called *model error* is large indeed. This points out the need for reliable information concerning the models — if the assumed model is very poor, we cannot be assured that the classifier we derive is the best, even among our model set. We shall return to the problem of choosing among candidate models in Chap. ??.

SAMPLE
COVARIANCE

ABSOLUTELY
UNBIASED

ASYMPTOT-
ICALLY
UNBIASED

3.3 Bayesian estimation

We now consider the Bayesian estimation or Bayesian learning approach to pattern classification problems. Although the answers we get by this method will generally be nearly identical to those obtained by maximum likelihood, there is a conceptual difference: whereas in maximum likelihood methods we view the true parameter vector we seek, $\boldsymbol{\theta}$, to be fixed, in Bayesian learning we consider $\boldsymbol{\theta}$ to be a random variable, and training data allows us to convert a distribution on this variable into a posterior probability density.

3.3.1 The Class-Conditional Densities

The computation of the posterior probabilities $P(\omega_i|\mathbf{x})$ lies at the heart of Bayesian classification. Bayes' formula allows us to compute these probabilities from the prior probabilities $P(\omega_i)$ and the class-conditional densities $p(\mathbf{x}|\omega_i)$, but how can we proceed when these quantities are unknown? The general answer to this question is that the best we can do is to compute $P(\omega_i|\mathbf{x})$ using all of the information at our disposal. Part of this information might be prior knowledge, such as knowledge of the functional forms for unknown densities and ranges for the values of unknown parameters. Part of this information might reside in a set of training samples. If we again let $\mathcal{D}$ denote the set of samples, then we can emphasize the role of the samples by saying that our goal is to compute the posterior probabilities $P(\omega_i|\mathbf{x}, \mathcal{D})$. From these probabilities we can obtain the Bayes classifier.

Given the sample $\mathcal{D}$, Bayes' formula then becomes

$$P(\omega_i|\mathbf{x}, \mathcal{D}) = \frac{p(\mathbf{x}|\omega_i, \mathcal{D})P(\omega_i|\mathcal{D})}{\sum_{j=1}^c p(\mathbf{x}|\omega_j, \mathcal{D})P(\omega_j|\mathcal{D})}. \quad (23)$$

As this equation suggests, we can use the information provided by the training samples to help determine both the class-conditional densities and the a priori probabilities.

Although we could maintain this generality, we shall henceforth assume that the true values of the a priori probabilities are known or obtainable from a trivial calculation; thus we substitute $P(\omega_i) = P(\omega_i|\mathcal{D})$. Furthermore, since we are treating the supervised case, we can separate the training samples by class into c subsets $\mathcal{D}_1, \dots, \mathcal{D}_c$, with the samples in $\mathcal{D}_i$ belonging to ω_i . As we mentioned when addressing maximum likelihood methods, in most cases of interest (and in all of the cases we shall consider), the samples in $\mathcal{D}_i$ have no influence on $p(\mathbf{x}|\omega_j, \mathcal{D})$ if $i \neq j$. This has two simplifying consequences. First, it allows us to work with each class separately, using only the samples in $\mathcal{D}_i$ to determine $p(\mathbf{x}|\omega_i, \mathcal{D})$. Used in conjunction with our assumption that the prior probabilities are known, this allows us to write Eq. 23 as

$$P(\omega_i|\mathbf{x}, \mathcal{D}) = \frac{p(\mathbf{x}|\omega_i, \mathcal{D}_i)P(\omega_i)}{\sum_{j=1}^c p(\mathbf{x}|\omega_j, \mathcal{D}_j)P(\omega_j)}. \quad (24)$$

Second, because each class can be treated independently, we can dispense with needless class distinctions and simplify our notation. In essence, we have c separate problems of the following form: use a set $\mathcal{D}$ of samples drawn independently according to the fixed but unknown probability distribution $p(\mathbf{x})$ to determine $p(\mathbf{x}|\mathcal{D})$. This is the central problem of Bayesian learning.

3.3.2 The Parameter Distribution

Although the desired probability density $p(\mathbf{x})$ is unknown, we assume that it has a known parametric form. The only thing assumed unknown is the value of a parameter vector $\boldsymbol{\theta}$. We shall express the fact that $p(\mathbf{x})$ is unknown but has known parametric form by saying that the function $p(\mathbf{x}|\boldsymbol{\theta})$ is completely known. Any information we might have about $\boldsymbol{\theta}$ prior to observing the samples is assumed to be contained in a *known* prior density $p(\boldsymbol{\theta})$. Observation of the samples converts this to a posterior density $p(\boldsymbol{\theta}|\mathcal{D})$, which, we hope, is sharply peaked about the true value of $\boldsymbol{\theta}$.

Note that we are changing our supervised learning problem into an unsupervised density estimation problem. To this end, our basic goal is to compute $p(\mathbf{x}|\mathcal{D})$, which is as close as we can come to obtaining the unknown $p(\mathbf{x})$. We do this by integrating the joint density $p(\mathbf{x}, \boldsymbol{\theta}|\mathcal{D})$ over $\boldsymbol{\theta}$. That is,

$$p(\mathbf{x}|\mathcal{D}) = \int p(\mathbf{x}, \boldsymbol{\theta}|\mathcal{D}) d\boldsymbol{\theta}, \quad (25)$$

where the integration extends over the entire parameter space. Now as discussed in Problem 12 we can write $p(\mathbf{x}, \boldsymbol{\theta}|\mathcal{D})$ as the product $p(\mathbf{x}|\boldsymbol{\theta}, \mathcal{D})p(\boldsymbol{\theta}|\mathcal{D})$. Since the selection of $\mathbf{x}$ and that of the training samples in $\mathcal{D}$ is done independently, the first factor is merely $p(\mathbf{x}|\boldsymbol{\theta})$. That is, the distribution of $\mathbf{x}$ is known completely once we know the value of the parameter vector. Thus, Eq. 25 can be rewritten as

$$\boxed{p(\mathbf{x}|\mathcal{D}) = \int p(\mathbf{x}|\boldsymbol{\theta})p(\boldsymbol{\theta}|\mathcal{D}) d\boldsymbol{\theta}.} \quad (26)$$

This key equation links the desired class-conditional density $p(\mathbf{x}|\mathcal{D})$ to the posterior density $p(\boldsymbol{\theta}|\mathcal{D})$ for the unknown parameter vector. If $p(\boldsymbol{\theta}|\mathcal{D})$ peaks very sharply about some value $\hat{\boldsymbol{\theta}}$, we obtain $p(\mathbf{x}|\mathcal{D}) \simeq p(\mathbf{x}|\hat{\boldsymbol{\theta}})$, i.e., the result we would obtain by substituting the estimate $\hat{\boldsymbol{\theta}}$ for the true parameter vector. This result rests on the assumption that $p(\mathbf{x}|\boldsymbol{\theta})$ is smooth, and that the tails of the integral are not important. These conditions are typically but not invariably the case, as we shall see in Sect. ???. In general, if we are less certain about the exact value of $\boldsymbol{\theta}$, this equation directs us to average $p(\mathbf{x}|\boldsymbol{\theta})$ over the possible values of $\boldsymbol{\theta}$. Thus, when the unknown densities have a known parametric form, the samples exert their influence on $p(\mathbf{x}|\mathcal{D})$ through the posterior density $p(\boldsymbol{\theta}|\mathcal{D})$. We should also point out that in practice, the integration in Eq. 26 is often performed numerically, for instance by Monte-Carlo simulation.

3.4 Bayesian Parameter Estimation: Gaussian Case

In this section we use Bayesian estimation techniques to calculate the a posteriori density $p(\boldsymbol{\theta}|\mathcal{D})$ and the desired probability density $p(\mathbf{x}|\mathcal{D})$ for the case where $p(\mathbf{x}|\boldsymbol{\mu}) \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$.

3.4.1 The Univariate Case: $p(\mu|\mathcal{D})$

Consider the case where $\boldsymbol{\mu}$ is the only unknown parameter. For simplicity we treat first the univariate case, i.e.,

$$p(x|\mu) \sim N(\mu, \sigma^2), \quad (27)$$

where the only unknown quantity is the mean μ . We assume that whatever prior knowledge we might have about μ can be expressed by a *known* prior density $p(\mu)$. Later we shall make the further assumption that

$$p(\mu) \sim N(\mu_0, \sigma_0^2), \quad (28)$$

where both μ_0 and σ_0^2 are known. Roughly speaking, μ_0 represents our best a priori guess for μ , and σ_0^2 measures our uncertainty about this guess. The assumption that the prior distribution for μ is normal will simplify the subsequent mathematics.

However, the crucial assumption is not so much that the prior distribution for μ is normal, but that it is known.

Having selected the a priori density for μ , we can view the situation as follows. Imagine that a value is drawn for μ from a population governed by the probability law $p(\mu)$. Once this value is drawn, it becomes the true value of μ and completely determines the density for x . Suppose now that n samples $x_1, \dots, x_n$ are independently drawn from the resulting population. Letting $\mathcal{D} = \{x_1, \dots, x_n\}$, we use Bayes' formula to obtain

$$\begin{aligned} p(\mu|\mathcal{D}) &= \frac{p(\mathcal{D}|\mu)p(\mu)}{\int p(\mathcal{D}|\mu)p(\mu) d\mu} \\ &= \alpha \prod_{k=1}^n p(x_k|\mu)p(\mu), \end{aligned} \quad (29)$$

where α is a normalization factor that depends on $\mathcal{D}$ but is independent of μ . This equation shows how the observation of a set of training samples affects our ideas about the true value of μ ; it relates the prior density $p(\mu)$ to an a posteriori density $p(\mu|\mathcal{D})$. Since $p(x_k|\mu) \sim N(\mu, \sigma^2)$ and $p(\mu) \sim N(\mu_0, \sigma_0^2)$, we have

$$\begin{aligned} p(\mu|\mathcal{D}) &= \alpha \prod_{k=1}^n \overbrace{\frac{1}{\sqrt{2\pi}\sigma} \exp \left[-\frac{1}{2} \left(\frac{x_k - \mu}{\sigma} \right)^2 \right]}^{p(x_k|\mu)} \overbrace{\frac{1}{\sqrt{2\pi}\sigma_0} \exp \left[-\frac{1}{2} \left(\frac{\mu - \mu_0}{\sigma_0} \right)^2 \right]}^{p(\mu)} \\ &= \alpha' \exp \left[-\frac{1}{2} \left(\sum_{k=1}^n \left(\frac{\mu - x_k}{\sigma} \right)^2 + \left(\frac{\mu - \mu_0}{\sigma_0} \right)^2 \right) \right] \\ &= \alpha'' \exp \left[-\frac{1}{2} \left[\left(\frac{n}{\sigma^2} + \frac{1}{\sigma_0^2} \right) \mu^2 - 2 \left(\frac{1}{\sigma^2} \sum_{k=1}^n x_k + \frac{\mu_0}{\sigma_0^2} \right) \mu \right] \right], \end{aligned} \quad (30)$$

where factors that do not depend on μ have been absorbed into the constants α , α' , and α'' . Thus, $p(\mu|\mathcal{D})$ is an exponential function of a quadratic function of μ , i.e., is again a normal density. Since this is true for any number of training samples, $p(\mu|\mathcal{D})$ remains normal as the number n of samples is increased, and $p(\mu|\mathcal{D})$ is said to be a *reproducing density* and $p(\mu)$ is said to be a *conjugate prior*. If we write $p(\mu|\mathcal{D}) \sim N(\mu_n, \sigma_n^2)$, then μ_n and σ_n^2 can be found by equating coefficients in Eq. 30 with corresponding coefficients in the generic Gaussian of the form

$$p(\mu|\mathcal{D}) = \frac{1}{\sqrt{2\pi}\sigma_n} \exp \left[-\frac{1}{2} \left(\frac{\mu - \mu_n}{\sigma_n} \right)^2 \right]. \quad (31)$$

Identifying coefficients in this way yields

$$\frac{1}{\sigma_n^2} = \frac{n}{\sigma^2} + \frac{1}{\sigma_0^2} \quad (32)$$

and

$$\frac{\mu_n}{\sigma_n^2} = \frac{n}{\sigma^2} \bar{x}_n + \frac{\mu_0}{\sigma_0^2}, \quad (33)$$

where $\bar{x}_n$ is the sample mean

$$\bar{x}_n = \frac{1}{n} \sum_{k=1}^n x_k. \quad (34)$$

We solve explicitly for μ_n and σ_n^2 and obtain

$$\mu_n = \left(\frac{n\sigma_0^2}{n\sigma_0^2 + \sigma^2} \right) \bar{x}_n + \frac{\sigma^2}{n\sigma_0^2 + \sigma^2} \mu_0 \quad (35)$$

and

$$\sigma_n^2 = \frac{\sigma_0^2 \sigma^2}{n\sigma_0^2 + \sigma^2}. \quad (36)$$

These equations show how the prior information is combined with the empirical information in the samples to obtain the a posteriori density $p(\mu|\mathcal{D})$. Roughly speaking, μ_n represents our best guess for μ after observing n samples, and σ_n^2 measures our uncertainty about this guess. Since σ_n^2 decreases monotonically with n — approaching σ^2/n as n approaches infinity — each additional observation decreases our uncertainty about the true value of μ . As n increases, $p(\mu|\mathcal{D})$ becomes more and more sharply peaked, approaching a Dirac delta function as n approaches infinity. This behavior is commonly known as *Bayesian learning* (Fig. 3.2).

BAYESIAN
LEARNING

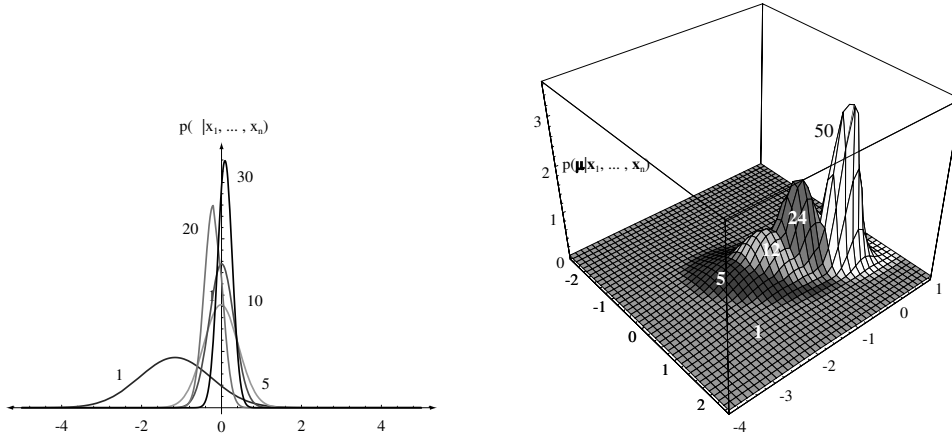


Figure 3.2: Bayesian learning of the mean of normal distributions in one and two dimensions. The posterior distribution estimates are labelled by the number of training samples used in the estimation.

In general, μ_n is a linear combination of $\bar{x}_n$ and μ_0 , with coefficients that are non-negative and sum to one. Thus μ_n always lies somewhere between $\bar{x}_n$ and μ_0 . If $\sigma \neq 0$, μ_n approaches the sample mean as n approaches infinity. If $\sigma_0 = 0$, we have a degenerate case in which our a priori certainty that $\mu = \mu_0$ is so strong that no number of observations can change our opinion. At the other extreme, if $\sigma_0 \gg \sigma$, we are so uncertain about our a priori guess that we take $\mu_n = \bar{x}_n$, using only the samples to estimate μ . In general, the relative balance between prior knowledge and empirical data is set by the ratio of σ^2 to σ_0^2 , which is sometimes called the *dogmatism*. If the dogmatism is not infinite, after enough samples are taken the exact values assumed for μ_0 and σ_0^2 will be unimportant, and μ_n will converge to the sample mean.

DOGMATISM

3.4.2 The Univariate Case: $p(x|\mathcal{D})$

Having obtained the a posteriori density for the mean, $p(\mu|\mathcal{D})$, all that remains is to obtain the “class-conditional” density for $p(x|\mathcal{D})$.^{*} From Eqs. 26, 27 & 31 we have

$$\begin{aligned} p(x|\mathcal{D}) &= \int p(x|\mu)p(\mu|\mathcal{D}) d\mu \\ &= \int \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right] \frac{1}{\sqrt{2\pi}\sigma_n} \exp\left[-\frac{1}{2}\left(\frac{\mu-\mu_n}{\sigma_n}\right)^2\right] d\mu \\ &= \frac{1}{2\pi\sigma\sigma_n} \exp\left[-\frac{1}{2}\frac{(x-\mu_n)^2}{\sigma^2+\sigma_n^2}\right] f(\sigma, \sigma_n), \end{aligned} \quad (37)$$

where

$$f(\sigma, \sigma_n) = \int \exp\left[-\frac{1}{2}\frac{\sigma^2+\sigma_n^2}{\sigma^2\sigma_n^2}\left(\mu - \frac{\sigma_n^2x + \sigma^2\mu_n}{\sigma^2+\sigma_n^2}\right)^2\right] d\mu.$$

That is, as a function of x , $p(x|\mathcal{D})$ is proportional to $\exp[-(1/2)(x-\mu_n)^2/(\sigma^2+\sigma_n^2)]$, and hence $p(x|\mathcal{D})$ is normally distributed with mean μ_n and variance $\sigma^2+\sigma_n^2$:

$$p(x|\mathcal{D}) \sim N(\mu_n, \sigma^2 + \sigma_n^2). \quad (38)$$

In other words, to obtain the class-conditional density $p(x|\mathcal{D})$, whose parametric form is known to be $p(x|\mu) \sim N(\mu, \sigma^2)$, we merely replace μ by μ_n and σ^2 by $\sigma^2 + \sigma_n^2$. In effect, the conditional mean μ_n is treated as if it were the true mean, and the known variance is increased to account for the additional uncertainty in x resulting from our lack of exact knowledge of the mean μ . This, then, is our final result: the density $p(x|\mathcal{D})$ is the desired class-conditional density $p(x|\omega_j, \mathcal{D}_j)$, and together with the prior probabilities $P(\omega_j)$ it gives us the probabilistic information needed to design the classifier. This is in contrast to maximum likelihood methods that only make point estimates for $\hat{\mu}$ and $\hat{\sigma}^2$, rather than estimate a *distribution* for $p(x|\mathcal{D})$.

3.4.3 The Multivariate Case

The treatment of the multivariate case in which Σ is known but μ is not, is a direct generalization of the univariate case. For this reason we shall only sketch the derivation. As before, we assume that

$$p(\mathbf{x}|\mu) \sim N(\mu, \Sigma) \quad \text{and} \quad p(\mu) \sim N(\mu_0, \Sigma_0), \quad (39)$$

where Σ , Σ_0 , and μ_0 are assumed to be known. After observing a set $\mathcal{D}$ of n independent samples $\mathbf{x}_1, \dots, \mathbf{x}_n$, we use Bayes' formula to obtain

$$\begin{aligned} p(\mu|\mathcal{D}) &= \alpha \prod_{k=1}^n p(\mathbf{x}_k|\mu)p(\mu) \\ &= \alpha' \exp\left[-\frac{1}{2}\left(\mu^t(n\Sigma^{-1} + \Sigma_0^{-1})\mu - 2\mu^t\left(\Sigma^{-1}\sum_{k=1}^n \mathbf{x}_k + \Sigma_0^{-1}\mu_0\right)\right)\right], \end{aligned} \quad (40)$$

^{*} Recall that for simplicity we dropped class distinctions, but that all samples here come from the same class, say ω_i , and hence $p(\mathbf{x}|\mathcal{D})$ is really $p(\mathbf{x}|\omega_i, \mathcal{D}_i)$.

which has the form

$$p(\boldsymbol{\mu}|\mathcal{D}) = \alpha'' \exp \left[-\frac{1}{2}(\boldsymbol{\mu} - \boldsymbol{\mu}_n)^t \boldsymbol{\Sigma}_n^{-1} (\boldsymbol{\mu} - \boldsymbol{\mu}_n) \right]. \quad (41)$$

Thus, $p(\boldsymbol{\mu}|\mathcal{D}) \sim N(\boldsymbol{\mu}_n, \boldsymbol{\Sigma}_n)$, and once again we have a reproducing density. Equating coefficients, we obtain the analogs of Eqs. 35 & 36,

$$\boldsymbol{\Sigma}_n^{-1} = n\boldsymbol{\Sigma}^{-1} + \boldsymbol{\Sigma}_0^{-1} \quad (42)$$

and

$$\boldsymbol{\Sigma}_n^{-1} \boldsymbol{\mu}_n = n\boldsymbol{\Sigma}^{-1} \hat{\boldsymbol{\mu}}_n + \boldsymbol{\Sigma}_0^{-1} \boldsymbol{\mu}_0, \quad (43)$$

where $\hat{\boldsymbol{\mu}}_n$ is the sample mean

$$\hat{\boldsymbol{\mu}}_n = \frac{1}{n} \sum_{k=1}^n \mathbf{x}_k. \quad (44)$$

The solution of these equations for $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}_n$ is simplified by knowledge of the matrix identity

$$(\mathbf{A}^{-1} + \mathbf{B}^{-1})^{-1} = \mathbf{A}(\mathbf{A} + \mathbf{B})^{-1} \mathbf{B} = \mathbf{B}(\mathbf{A} + \mathbf{B})^{-1} \mathbf{A}, \quad (45)$$

which is valid for any pair of nonsingular, d -by- d matrices $\mathbf{A}$ and $\mathbf{B}$. After a little manipulation (Problem 16), we obtain the final results:

$$\boldsymbol{\mu}_n = \boldsymbol{\Sigma}_0 \left(\boldsymbol{\Sigma}_0 + \frac{1}{n} \boldsymbol{\Sigma} \right)^{-1} \hat{\boldsymbol{\mu}}_n + \frac{1}{n} \boldsymbol{\Sigma} \left(\boldsymbol{\Sigma}_0 + \frac{1}{n} \boldsymbol{\Sigma} \right)^{-1} \boldsymbol{\mu}_0 \quad (46)$$

(which, as in the univariate case, is a linear combination of $\hat{\boldsymbol{\mu}}_n$ and $\boldsymbol{\mu}_0$) and

$$\boldsymbol{\Sigma}_n = \boldsymbol{\Sigma}_0 \left(\boldsymbol{\Sigma}_0 + \frac{1}{n} \boldsymbol{\Sigma} \right)^{-1} \frac{1}{n} \boldsymbol{\Sigma}. \quad (47)$$

The proof that $p(\mathbf{x}|\mathcal{D}) \sim N(\boldsymbol{\mu}_n, \boldsymbol{\Sigma} + \boldsymbol{\Sigma}_n)$ can be obtained as before by performing the integration

$$p(\mathbf{x}|\mathcal{D}) = \int p(\mathbf{x}|\boldsymbol{\mu}) p(\boldsymbol{\mu}|\mathcal{D}) d\boldsymbol{\mu}. \quad (48)$$

However, this result can be obtained with less effort by observing that $\mathbf{x}$ can be viewed as the sum of two mutually independent random variables, a random vector $\boldsymbol{\mu}$ with $p(\boldsymbol{\mu}|\mathcal{D}) \sim N(\boldsymbol{\mu}_n, \boldsymbol{\Sigma}_n)$ and an independent random vector $\mathbf{y}$ with $p(\mathbf{y}) \sim N(\mathbf{0}, \boldsymbol{\Sigma})$. Since the sum of two independent, normally distributed vectors is again a normally distributed vector whose mean is the sum of the means and whose covariance matrix is the sum of the covariance matrices (Chap. ?? Problem ??), we have

$$p(\mathbf{x}|\mathcal{D}) \sim N(\boldsymbol{\mu}_n, \boldsymbol{\Sigma} + \boldsymbol{\Sigma}_n), \quad (49)$$

and the generalization is complete.

3.5 Bayesian Parameter Estimation: General Theory

We have just seen how the Bayesian approach can be used to obtain the desired density $p(\mathbf{x}|\mathcal{D})$ in a special case — the multivariate Gaussian. This approach can be generalized to apply to any situation in which the unknown density can be parameterized. The basic assumptions are summarized as follows:

- The form of the density $p(\mathbf{x}|\boldsymbol{\theta})$ is assumed to be known, but the value of the parameter vector $\boldsymbol{\theta}$ is not known exactly.
- Our initial knowledge about $\boldsymbol{\theta}$ is assumed to be contained in a known a priori density $p(\boldsymbol{\theta})$.
- The rest of our knowledge about $\boldsymbol{\theta}$ is contained in a set $\mathcal{D}$ of n samples $\mathbf{x}_1, \dots, \mathbf{x}_n$ drawn independently according to the unknown probability density $p(\mathbf{x})$.

The basic problem is to compute the posterior density $p(\boldsymbol{\theta}|\mathcal{D})$, since from this we can use Eq. 26 to compute $p(\mathbf{x}|\mathcal{D})$:

$$p(\mathbf{x}|\mathcal{D}) = \int p(\mathbf{x}|\boldsymbol{\theta})p(\boldsymbol{\theta}|\mathcal{D}) d\boldsymbol{\theta}. \quad (50)$$

By Bayes' formula we have

$$p(\boldsymbol{\theta}|\mathcal{D}) = \frac{p(\mathcal{D}|\boldsymbol{\theta})p(\boldsymbol{\theta})}{\int p(\mathcal{D}|\boldsymbol{\theta})p(\boldsymbol{\theta}) d\boldsymbol{\theta}}, \quad (51)$$

and by the independence assumption

$$p(\mathcal{D}|\boldsymbol{\theta}) = \prod_{k=1}^n p(\mathbf{x}_k|\boldsymbol{\theta}). \quad (52)$$

This constitutes the solution to the problem, and Eqs. 51 & 52 illuminate its relation to the maximum likelihood solution. Suppose that $p(\mathcal{D}|\boldsymbol{\theta})$ reaches a sharp peak at $\boldsymbol{\theta} = \hat{\boldsymbol{\theta}}$. If the prior density $p(\boldsymbol{\theta})$ is not zero at $\boldsymbol{\theta} = \hat{\boldsymbol{\theta}}$ and does not change much in the surrounding neighborhood, then $p(\boldsymbol{\theta}|\mathcal{D})$ also peaks at that point. Thus, Eq. 26 shows that $p(\mathbf{x}|\mathcal{D})$ will be approximately $p(\mathbf{x}|\hat{\boldsymbol{\theta}})$, the result one would obtain by using the maximum likelihood estimate as if it were the true value. If the peak of $p(\mathcal{D}|\boldsymbol{\theta})$ is very sharp, then the influence of prior information on the uncertainty in the true value of $\boldsymbol{\theta}$ can be ignored. In this and even the more general case, though, the Bayesian solution tells us how to use *all* the available information to compute the desired density $p(\mathbf{x}|\mathcal{D})$.

While we have obtained the formal Bayesian solution to the problem, a number of interesting questions remain. One concerns the difficulty of carrying out these computations. Another concerns the convergence of $p(\mathbf{x}|\mathcal{D})$ to $p(\mathbf{x})$. We shall discuss the matter of convergence briefly, and later turn to the computational question.

To indicate explicitly the number of samples in a set for a single category, we shall write $\mathcal{D}^n = \{\mathbf{x}_1, \dots, \mathbf{x}_n\}$. Then from Eq. 52, if $n > 1$

$$p(\mathcal{D}^n|\boldsymbol{\theta}) = p(\mathbf{x}_n|\boldsymbol{\theta})p(\mathcal{D}^{n-1}|\boldsymbol{\theta}). \quad (53)$$

Substituting this in Eq. 51 and using Bayes' formula, we see that the posterior density satisfies the recursion relation